Cloud Computing — DeployX storage notes

Object storage (Documents)

- Buckets and keys instead of directories; served over HTTP at `/files/<key>`
- Versioning keeps every upload; a lifecycle rule expires old versions after 7 days
- Erasure coding splits each object into data + parity shards across 4 drives, so losing a drive loses no data
- Presigned URLs give time-limited upload/download without sharing credentials

Block storage (Volumes)

- `deploy.volumes` mounts a named Docker volume at `/data`
- Storage is separate from compute: it outlives containers, redeploys and blue/green switches

Virtualized compute (Resource limits)

- `deploy.resources` → Docker `--cpus`, `--memory`, `--pids-limit` → Linux cgroups
- The container sees a hard cap (0.5 CPU, 256 MB, 128 processes) regardless of the host

Placement and security

- `driver: auto` brokers the deployment target (docker_local, Railway, Fly.io)
- Trivy scans the image before deploy; HIGH/CRITICAL findings block it